



Estimating the lives that could be saved by expanded access to weight-loss drugs

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Obesity is a major public health crisis in the United States (US) affecting 42% of the population, exacerbating a spectrum of other diseases and contributing significantly to morbidity and mortality overall. Recent advances in pharmaceutical interventions, particularly glucagon-like peptide-1 (GLP-1) receptor agonists (e.g., semaglutide, liraglutide) and dual gastric inhibitory polypeptide and GLP-1 receptor agonists (e.g., tirzepatide), have shown remarkable efficacy in weight-loss. However, limited access to these medications due to high costs and insurance coverage issues restricts their utility in mitigating the obesity epidemic. We quantify the annual mortality burden directly attributable to limited access to these medications in the US. By integrating hazard ratios of mortality across body mass index categories with current obesity prevalence data, combined with healthcare access, willingness to take the medication, and observed adherence to and efficacy of the medications, we estimate the impact of making these medications accessible to all those eligible. Specifically, we project that with expanded access, over 42,000 deaths could be averted annually, including more than 11,000 deaths among people with type 2 diabetes. These findings underscore the urgent need to address barriers to access and highlight the transformative public health impact that could be achieved by expanding access to these novel treatments.

obesity | diabetes | mortality | weight-loss drugs | drug access

Obesity remains a formidable public health crisis in the United States (US), contributing substantially to both morbidity and mortality (1). It is a major risk factor for various chronic diseases, including cardiovascular disease, type 2 diabetes, and myriad forms of cancer. For example, the risk of endometrial cancer in people with severe obesity is seven times higher compared with the rest of the population who are not overweight (2). Among many other causes of mortality and morbidity, obesity is a risk factor for poor medical outcomes from a plethora of infectious diseases, surgical-site infections, and nosocomial infections (3). Compounding the challenge, the obesity prevalence in the US is skewed toward the vulnerable populations, including those with lower socioeconomic status and limited insurance coverage (4). Adults with obesity are likely to spend over \$1,800 more in medical costs per year than those without obesity (5). Altogether, the economic burden of obesity is immense, with estimated annual medical costs exceeding \$170 billion (5).

Historically, the development of pharmaceutical interventions for weight-loss has been fraught with challenges, including safety concerns and limited efficacy (6). However, a paradigm shift has occurred with the discovery that drugs originally intended for diabetes management, particularly glucagon-like peptide-1 (GLP-1) and dual gastric inhibitory polypeptide and GLP-1 (GIP/GLP-1) receptor agonists, can induce substantial weight-loss (7). These next-generation medications, such as semaglutide, liraglutide, and tirzepatide, have demonstrated remarkable efficacy in clinical trials, resulting in weight reductions previously considered unattainable through pharmaceutical means alone (7). As a result, these medications are increasingly gaining approval from the US Food and Drug Administration (FDA) for the treatment of obesity, ushering in a new era in obesity management (8).

Semaglutide, initially approved in 2017 for type 2 diabetes as Ozempic, expanded its use in 2021 with FDA approval as Wegovy for chronic weight management. This approval is for individuals aged 12 and older with a body mass index (BMI) of 30 kg/m² or higher or a BMI between 27 to 30 kg/m² with at least one weight-related condition, such as hypertension, type 2 diabetes, or high cholesterol. Tirzepatide, another medication in this class, received approval in 2022 for type 2 diabetes as Mounjaro and most recently in November 2023 as Zepbound for chronic weight management in adults with similar criteria to Wegovy.

Despite their promise, widespread access to these medications remains a critical barrier to addressing the escalating obesity epidemic in the US. While semaglutide (Ozempic, Wegovy) and tirzepatide (Mounjaro, Zepbound) have demonstrated efficacy in reducing body weight, their high-cost, limited supply, and lack of comprehensive insurance coverage

Significance

While obesity is rarely listed as the primary cause of death, it exacerbates a range of other chronic and acute diseases, overall leading to deaths that would not otherwise have occurred in the absence of obesity. This study quantifies the mortality impact of limited access to novel and highly effective weight-loss medications in the United States. By highlighting the potential number of lives lost due to restricted access, our research underscores the urgent need for policy interventions to improve affordability and insurance coverage. The findings have significant implications for public health strategies to combat the obesity epidemic and reduce associated mortality.

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The authors declare no competing interest.

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